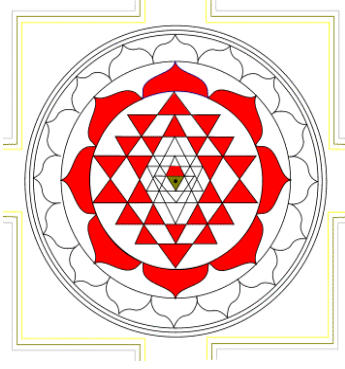




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By Niraj Kumar .

From the chief editors desk

A dynamic renovation perpetuated through every time fraction is the notion of agreeability -says a great poet Kalidas . Every new year another turn of the eternal circulation of the wheel of time generates bliss which a produce of syncing with transitory disposition of the universal phenomena . This explains the variety of tactics in the tantras . A great relief from the monotony of pursuing just one objective in just one particular way . Answering the query Why so many options to choose from ,when it comes to upayas khemendra says -this upaya suits this guy and that one suits another guy . The Agama give you many upayas to choose from besides the monotony of of one single upaya you also have the joy of trying out new new things . . . dūcunt hominēs per saecula Rōmam - all roads lead to Rome .

Expression as one minor upanishad puts it ,gets divergent with the gradations of separation between word and meaning . A perfect homogeneity of the both is truth and their separation ,heading towards falsehood which turns out to be a state of complete isolation of word and meaning unrelated from one another . The innate unison and the order of progression of word and it's meaning during the power and process of the universal phenomena ,which is evaluated by our antique canons in terms of names and forms -another synonyms for word and meaning .

Tantrikas rely more on their subjective experience rather than dialectic hairsplitting . The methodology of imparting a tantrik experience includes various diagramming semiotics and analogy all this to abridge the reciprocation of descriptive intrigue. L

All the organs of a beautiful peacock are latent in the egg of a pea hen . All that betel is splendour of blue, green white and purple hues of plumage,feathers tuft and beak etc. invisible but existent in the yolk of a pea hen egg lie dormant in a fluid of yellow and colourless jelly . All words and meaning or to be more precise the absolute convergence Of collective word I.e. nada and the absolute convergence of all corresponding meaning I.e. Bindu remain indistinguishable in a perfect homogeneous

blend in the para mode of expression . Para as such is believed to be inseparable due to a perpetual unison with Paranand the lord which out of his sheer will becomes the agency of creation sustenance etc and turns extrovert .

Para the transcendent aspect of vak or expression gets grosser and comes to explained by the analogy of a banyan seed ,it contains the potentiality of an entire. Baniya tree and thus the myriad multiplication of the trail of a tiny seed and gigantic tree. Pashyanti is so called because it predated and outlasts the causal state of deep dream less sleep and the ultimate dissolution . The same in an individual manifestation is called Madhyama The Intermediate vak and manifests as individual reason .in this mode all letters or rather the eight groups of letters are arranged like beans in a pod . Sheathed by the shell hibernate the eight groups of alphabet called the Matrika which bind a conditioned extrovert aspirant to his lower nature of lust,wrath ,greed,delusion,vanity,Mali a vice and virtue ,reduce a man to a biological being -a Pashu .as these unite with the experience of unison with one's shivahood they become their Shaktis and remain subservient .

The substratum or the origin and subsiding of these four modes of expression is in the pure subjective consciousness which dawns in the form of Swara . Swara literally a sound in this context refers not merely to the anahat nada that Mystics and yogins familiarise through various yogic techniques but the uninterrupted streamline of awareness that extends beyond the struck,unstruck and both struck as well as unstruck sounds . Pratha means to get substantiate . As the voices in the head and the mental jabbervockey subsides the innate spontaneous streamline of awareness gets louder and distinct .स्वरःप्रथते as one aphorism of vatulnath puts , can also be interpreted as above . The great mystic kabeer confines his experience as to how he went beyond the quagmire of intriguing voices in the head by holding the rope of Shabda.

हम तो बच गये साहिब दया से शबद डोर गहि उतरे पारा ।

By the munificence of our lord we held on to the rope of Shabda or nada and crossed the mire of objective phenomena .

Swami Paranand Tirth

Founder president CEO

YTA Yoga Tantra and Agama (inc) cal. USA

And Yoga Tantra et agama centre Quebec Canada .

Food for health



(Dr Vishwajeet Rohil MBBS,MD)

The present article we will be briefly discussing a detailed account of each of the nutrients that the human body requires for physical and spiritual well being .

Vegetarian v/s Non-Vegetarian:

As Sri Yukteswar Giri ji, guru of Sri Paramahansa Yogananda, states in his Holy Science, “Can flesh be considered the natural food of man, when both his eyes and his nose are so much against it, unless deceived by flavors of spices, salt and sugar. On the other hand, how delighted do we find the fragrance of fruits, the very sight of which often makes the mouth water.” You can ask a question to yourself, can you eat raw vegetables or raw fruits, answer will be yes but when you ask about eating raw meat, obviously the answer is a big ‘no’. Why, because it is not natural, the human nature itself will resist even such a thought.

The key element to the sattvic diet is vegetarianism. Yoga teaches that vegetarian diet is not only essential for the spiritual life, but is also the basis for good health. Many people worry that when they stop eating meat and fish, they might be in danger of some nutritional deficiency. This is not the case as all the nutrients you need can easily be obtained from a vegetarian diet. In fact research has shown that in many ways a vegetarian diet is healthier than that of a typical meat-eater. Research has shown not only do vegetarians live longer and have two to three times the endurance than meat-eaters, they are less subject to the health problems that meat-eaters experience. A vegetarian diet is known to confer a wide range of health benefits. Vegetarians are less prone to heart disease, hypertension, obesity, diabetes, various cancers, diverticular disease, bowel disorders, gall stones, kidney stones, and osteoporosis. The eating of flesh foods significantly increases major chronic diseases like heart disease, hypertension, cancer, kidney disease, and osteoporosis. People who eat meat are also at a higher risk of various viral, bacterial, fungal, and parasitical infections. Toxoplasmosis is one of these. There is now the possibility of Mad Cow Disease. Flesh food being at the top of the food chain has about 15 times more pesticides and herbicides than vegetable food. Even dairy products have about 5 times more pesticides and herbicides than vegetarian food.

Sri Ramana Maharshi was once asked what the most important aid to meditation was, he replied a pure vegetarian diet. He quoted the ancient Chandogya Upanishad D II 26.2 that says “when food is pure, the mind is pure, when the mind is pure, concentration is steady, when concentration is achieved one can loosen all the knots of the heart that bind us.” Vegetarianism is an essential constituent for purification of mind.

Flesh food increases the animal-like tendencies in the body such as the vibrations of anger, lust, fear and murderous impulses. The energy of a flesh food diet adds to the impurities of the mind and the nervous system. It communicates the energy of destruction to the cells and brings the energy of death into our auric fields reducing the flow of higher prana into the body. The lives of the creatures you have eaten weigh down your astral body with their negative feelings of fear at their time of death. Flesh food creates a tamasic (dull and heavy) effect on the physical body and mind. It clogs the channels of the subtle body, the 72,000 nadis through which the Kundalini needs to move freely to do its spiritualizing work, and tends to make the mind insensitive. Every animal that is slaughtered for human consumption brings the pain of death into your body. The animal is killed with violence. That violence causes the animal to experience very intense pain and there is a release of highly toxic chemicals into the blood circulation like adrenaline and other chemicals due to fear, reaction and pain, as it dies. That pain and highly toxic chemicals remains in the meat even after you've prepared and cooked it. When you eat the meat, then you eat pain and highly toxic chemicals along with it. That pain becomes lodged in your body, heart, and mind. That violence and pain which you consume will eat you also! The toxic chemicals are extremely harmful to human body, physically as well as mentally.

Buddhism is originally vegetarian and, according to the Dalai Lama, Buddhism should return to vegetarianism. The Mahaparinirvana Sutra sums it up, “The eating of meat extinguishes the seed of great compassion. A key component for creating a sattvic mind is the practice of Ahimsa. A vegetarian way of life actively creates six aspects of Ahimsa: (1) compassion and non-cruelty toward animals; (2) preserving the earth and its ecology; (3) feeding the hungry; (4) preserving human life; (5) preservation of personal health; (6) and inspiring peace.

Diet for Sadhak

Pure Satvik vegetarian diet is a prior requirement of almost all Sadhanas. All the "Taamsik" food like onion, garlic, spices etc. are totally banned for sadhaks. Basically you are what you

eat. Consumption of non-vegetarian food leads to negative thoughts of lust, violence etc. As purpose of Sadhana is also to get rid of the past bad karmas. Consumption of non-veg food is only going to increase the bad karmas. One requires full concentration of mind to meditate during Sadhana. Drinking, smoking or using tobacco in any other form or any other kind of addiction etc. impair mind and so such addictions are also prohibited for those practicing Sadhanas.

The equivalent of a traditional sattvic diet today consists of organic, whole, natural fruits, vegetables, nuts, seeds and grains. A modern sattvic diet emphasizes foods grown in harmony with nature by organic farmers, on good soils, ripened naturally and prepared with an attitude of love. Such foods carry the highest prana and consciousness. A modern sattvic diet does not include junk food, excessively spicy or salty foods, fried food, white flour, white sugar, and other forms of food that unnaturally stimulate your blood sugar or your mind. It avoids meat, fish and alcohol, and often garlic, onions and eggs as well. We should remember that the sattvic yogic diet is a special type of spiritual vegetarian diet aimed at helping us transcend body consciousness and calm the vrittis of the mind and increase a higher type of Prana. For this reason, the yogic diet is more disciplined than a regular vegetarian diet, generally lighter, and includes fasting and the taking of live-foods (uncooked roots, leafy vegetables and fruit, which are called specifically the food of the rishis or the food of the yogis). Raw foods are rich in prana, which brings the prana force not only into the body, but also into the mind helping to purify the 72,000 nadis. Raw foods are the most powerful foods for enhancing the flow of Kundalini energy and cleansing the nadis through which the Kundalini flows on the subtle plane. A yogic sattvic diet helps us become superconductors of prana. As the prana in our body increases, these nadis naturally open, affording us deeper perceptions of truth, reality and universality.

Out of our five sense organs, only the tongue is entrusted with two functions: speech and tasting of different flavors (Rasas).. It is said that if only the tongue is brought under control, then the control over all other sense organs becomes easy. Very tasty and spicy foods are usually consumed excessively. This will often produce several diseases of the stomach, intestines; liver and the tongue. In addition, sleep and laziness get enhanced and the means of God-realization like self-study, chanting prayers, meditation, devotional singing etc cannot be practiced with due earnestness or concentration. The mind remains predominantly occupied with the ideas and analysis of what is to be prepared, how, where, at what time and under what circumstances the food will be consumed and so on. We waste a lot of time, huge energy and incur great expense in the preparation of certain tasty dishes and the taste lasts only while the item is in the mouth. Once the item goes beyond the palate, it has only one final shape - it turns to stool or other waste of the body.

Nutrients required by human body

Nutrients are usually divided into five classes: carbohydrates, proteins, fats (including oil), vitamins and minerals. We also need fiber and water. All are equally important to our well being. It will be discussed in details in Part-II

Dr.Vishwajeet Rohil

Wisdom from shivas rattle drum .

WISDOM FROM SHIVA'S RATTLE DRUM. ...

THE DANCE OF SHIVA



A Tibetan mendicant shaman with shivas damaru -chod a ritual playing of the rattle drum

For severing the bonds of ego .

नृत्तावसाने नटराजराजो ननाद ढक्कां नवपञ्चवारम् ।

उद्धर्तुकामः सनकादिसिद्धानेतद्विमर्शे शिवसूत्रजालम् ॥१॥

And after his cosmic dance shiva played his damru fourteen times to redeem the four sons of brahma thus producing fourteen aphorisms on which the entire grammar of panini is caricatured .

The dance of shiva is in fact the dance of dissolution of the conventional phenomena that comes from an innate divergent perspective . The refinement comes from acquiring the world view from the revelations after their thorough study which in turns requires exactitude in expression I.e. grammar . This cosmic dance is a prelude to the termination of chaotic aberrant and naive perspective and the beginning of a refined cultured and sophisticated world view .

The apostles of this supreme wisdom and ethics based on it were the four off springs of Brahma the creator . On being asked to procreate they defied their father and became the disciples of shiva to pursue on the lines of pure metaphysical experience of the absolute truth .

The quagmire of downright naivety specially when it comes to expression is so much gripping that even the four enlighten boys ,the sons of brahma the creator are gripped in it . No verbal interaction or gestures can deliver one from the view point acquired by an accumulation of faulty and erroneous input of nescience to which an individual is subject since the beginning of this creation and till the wisdom of truth and righteousness dawns . There is an episode in the bhagavatam about the incompetence of brahma the creator in confining the supreme metaphysics

Social truth to his sons . A celestial swan incarnated and imparted the supreme truth to sanat Sanandan sanaktsujat and sanatkumar . Brahma the creator being too much engrossed in the relative notion of the objective phenomena that the queries of born mystics went above his head . It takes deep intuitive insight even to comprehend these let alone answering questions pertaining to them.

In this parlance lord shiva redeemed them and they were awakened by the fourteen beats of his rattle drum . The phonetic semblance lead to the evolution of one of most scholastic feats of the world ,the school of panini's grammar but en passant

The mystical interpretation of the aphorisms as they conventionally came to be called ,also has a great cache of eastern metaphysics encrypted in them .what a discursive scholar would ascertain after learning the grammar and then the revelations through it ,is already encrypted in the aphorisms . The author of kashika on paninis aphorism has elaborated the mystic part of these aphorisms .sutras as it were are like huge bunch of cotton spun into thread and winded into a reel . Somewhat like portable digital format files . A mystic approach has another benefit of developing a special faculty of scholastic intuition. What the author meant by certain aphorisms is not distinct and many commentators have come out various and conflicting view points regarding the purport of various aphorisms .beginning with a presupposition of the divine origin and authenticity of these can get an easy inlet for an aspirant into the intricacies of paninis ingenious inventions of the discipline of human expression.



From a gross view point .Lord Shiva is portrayed as freakish god with matted locks and draped in a pachyderm hide or leopard skin . He rides a bull and is signified by the trident and tambourine that he carries in his hands . These are rather semiotic depictions of various esoteric mystic principles .for example the bull denotes dharma or the elephant hide denotes annihilation of pride or vanity. Reptiles around his neck stand for his universal agreeability or theorem to assimilate even the vehement and hostile factions of the universe so forth and so on . Semiotics replaced lofty methodological expositions as the competence of the aspirants declined and it became necessary to introduce the cult among hoi polloi. Since the masses went by seeing rather than by studying or understanding . Due to the doctrinal vicissitude caused by plunderers and rampagers the only remnants of an illustrious institution of religion

which survived was the hotchpotch of too many relics and icons shrines - unarticulated ,deranged and almost devoid of any contemporary pragmatic import . Fortunately the lineage of spiritual experience of the sages of enlightenment wasn't interrupted and that lead to a fresh trend of esoteric evaluation of the glorious tradition that was passing into oblivion by and by .

The trident is a semiotic depiction of para vak or they're subjective consciousness unalloyed by any taint of pure subjective conditioning. This mode of vak is the stage of perfect homogeneity of the word and meaning . In the same way the rattle drum called kshudra dundubhi in Sanskrit and damru or dhakka in colloquial Sanskrit is a symbol of pashyanti vak or the grosser expression in the realm of the cosmic intelligence or महत्त . In this state word and meaning begin to separate and the mode is a bit divergent . ईषत्बहिर्मुख सुषुप्ति दशा साक्षी slightly divergent witnessing awareness of dreamless sleep and dissolution since from a metaphysical viewpoint deep sleep and dissolution are not two dissimilar experiences ,as shankarananda defines it in his narsimh tapini Upanishads deepika .

Pashyanti is also generated by striking percussionist instruments mounted by leather says natananandnath in his commentary on the Kamakala vilas of punyanand nath yati

Thus even shiva while playing the percussion is only an agent of the expression of the pashyahi state of vak .the eternal word in this stage is in a state of the unison of word and meaning .this surpasses the conditioning of empirical reason or the individual intellect .

वाग्धैव पश्यन्ती वदति .(मंडल ब्राह्मण) शतपथ ब्राह्मणम्

Brahman has some clues about this seeing mode of vak . Hearing can be related to word where as the meaning refers to seeing . From the stand point of the predominance of meaning vak is seeing I.e. meaning . In an oriental parlance truth is definable in terms of absolute convergence of collective word and meaning . The divergence of word and meaning is falsehood and their convergence , homogeneity and unison -the expression of truth .thus There are three trends one from the viewpoint of nada or word another from that of bindu or meaning and yet another from the stand point of their homogeneity

The last however pertains only to the revelation(offenerkeit)and revelation ability(offendurung)part of the phenomena .The theorem of a mystic adept shivanandnath the commentator of matrikachakraviveka which is a compendium of alphabetic configuration of the mystic diagram of shri chakra-

Can be interpreted in either way . नादः बिंदुकारणमिति भारतीयानां चव्यवहारः

Nada is the cause of bindu or vice versa is also a tradition among the bharatiyas inhabitants of bharatvarsa or those who maintain that bharati or the Sanskrit language is their lingua Franca .

The exposition conveys the esoteric specification of the mystic sounds of SHIVAS drum as he played it fourteen times . This encrypts the crux of entire eastern metaphysics specially when it comes to rendering the purport of the revelations.

The revelations so as to establish the dual nature of the power and process of creation . In the form of word and meaning .

Para vak or the unalloyed pure unconditioned awareness is ever in unison with parameshwara . A grosser manifestation called pashyanti is the mode that also manifests in the form of rhythm and beats produced while playing percussion like a drum or tambourine . These aphoristic beats are

As austere as the revelations of the Vedas and owing to their superhuman origin commonplace mortals cannot decipher their mystic import .

For a want of proper school of grammatical thought the entire gamut of Vedic scholarship was either being tainted or being construed to conflict the theory of divine origin of word . Specially the one that pertains to innate and spontaneous relation between the word and its meaning .

The discreet lineage of the spiritual truth and wisdom was getting obsolete due to an absence of exactitude in expression and thus the advancement of knowledge and religion .

The faculty of human expression and. Cognition was renovated by the supreme preceptor shiva not through any verbal communication but by playing his rattle drum 14 times . The entire edifice of paninis and patanjalis grammar is caricatured on these fourteen aphorisms which besides being the ingenious devices of encrypting and explaining grammatical principles also reinforces the supra mundane origin of the revelations and their auxiliaries . This entirely replaced the shabdbrahmvaad theory of the predecessors .it came to get so much popular that without the mediation of grammar the parroting of Vedic revelation came to be considered as inferior.

These aphorisms were not only grammatical encryptions they also contained immense spiritual wisdom pertaining to the power and process of I creation interlaced in them . In short agama substituted their The nigama counterpart which was established on the principle of the revelation as the ultimate testimony of evidence . The meaning part of it as it were came to get obscure and marred which is why we have so many interpretations of the Vedic dictum and the dogmas based on this multiplicity . These jarring sects have been philosophic horn locked buffaloes and focused only on refuting other doctrines and asserting their own . In this process the illustrious lineage of pure metaphysical experience was reduced to the ever conflicting institutions of religious discipline .

The agama, specially the one devised to assimilate the ultimate purport of the scriptures found it more consistent to begin with exactitude or the refinement of the word for attaining the ultimate meaning . Hence the supernatural origin or metaphysical authenticity of the fourteen syllables or the sutras which either due to phonetic semblance or to establish their roots in the supreme omniscient being are the beats of the rattle drum .

The term sutra which is popularly translated as an aphorism actually means thread an enormous quantity of knowledge can be spun into thread and winded into a small portable reel that can be carried in the pocket . This aphoristic style owes a lot to the encryption and ontological formulation for the specifications of conceptualisation

For reuse etc. this has been the most conventional mode of the propagation of knowledge among the Buddhists we'll as the Hindus

Some scholars interject here that these en passant syllables bearing a semblance to the beats of a rattle drum have been deliberately construed to create an impression as if the source of the most efficacious school of grammar is encrypted in them .

As we have already mentioned the arcane lore of the agamic tradition has been deliberately designed to include a man of considerable intellectual competence as well the hoi polloi .so the laities bull the vehicle of The Lord is actually the semiotic depiction of righteousness or religion .

Nandi is dharma and it is only from this righteous stand point that one can interpret the purport of the revelations and what ever else is contributory to them .

The following esoteric as well exoteric rendering of the fourteen holy formulae is attributed to nandikeshwar the celestial bull who in turn is the semiotic depiction of dharma or righteousness to be more precise the principle of righteousness. The exactitude of speech and righteousness go hand in hand .exactitude n expression means heading the truth . From a semiotic perspective truth or to be more precise the metaphysical truth rides dharma or righteousness . Th vehicle of truth I.e. righteousness ,personified as the celestial Bull is the most authentic person to interpret the arcane encryptions to commoners . Hence the epistemological study cannot fructify without an admixture of theistic credulousness .

Paranand swami is the founder president and CEO

YTA Yoga Tantra & Agama (inc)

KAL CHAKRA AND TIME AWARENESS FROM A NEUROLOGICAL PERSPECTIVE .

Neeraj kumar .

Neurons in Human Brain

A neuron has thousands of branches called dendrites. The head and tail both sides have these branching. Then, there are synapses on these dendrites through which the ions are transported and information exchanged. There are 1000-10000 synapses per neuron in human brain. On an average, this number is 7000 synapses per neuron. Synapses are the eyes of neurons through which information is exchanged. The description of Puruṣa in the Veda fits well with structure of neuron. Interestingly, it is these neurons which mediate active consciousness. Can we decipher the intuitive flash of seers through the Puruṣa-Sukta further?



Image 4: Structure of Neuron

Seat of Creative Consciousness: Billions of Neurons

The first verse of the Puruṣa –Sukta gives four numbers- thousand, thousand, thousand and ten. When we multiply all these, we get the number 1010 which is 10 billion. The total number of neurons estimated in human brain is 86 billion. There are 1 billion neurons in spinal cord, 69 billion in cerebellum and 16 billion in cerebral cortex. Recent advances in neuroscience has clearly established that cerebrum is the seat of dynamic consciousness.

Few years back, Nobel Prize winner biologist for discovery of double helix structure of DNA, Francis Crick with Christof Koch(2007) advanced argument for neuronal correlate of consciousness(NCC) defined as minimal neuronal mechanism jointly sufficient for any specific conscious experience. Since then search for “consciousness neurons” has become a challenging exploration.

Even if spinal cord is severed, victims are merely paralyzed in legs, arms and torso and without body sensation. But. They continue to be conscious and experience life in variegated

forms. Similarly, even if cerebellum which is involved with motor control, posture, and which possess highest number of neurons is found to be not necessary for conscious experience. Even if the cerebellum gets damaged, a patient does not lose consciousness. Cerebellum is not involved in subjective experience. The conscious experience arises only when neurons in cerebral cortex is excited. Cerebral cortex is the largest part of the brain. There are two hemispheres on right and left joined by a bundle of fibres, corpus callosum, that carries information between the two hemispheres. Each hemisphere has four lobes-frontal, parietal, temporal(side) and occipital(back side). Thus, the cerebrum comprises of 8 lobes containing a total of 16 billion neurons.

The number interpreted from the first verse of the Puruṣa –Sukta, i.e. 10 billion is very close to 16 billion.

Since the set of 16 billion neurons is the seat of active consciousness, does the cortical neurons relate to the outer world in anyway?

Cortical Neurons and Cosmic History

There are 100 billion galaxies estimated to be scattered in our universe. We again find that our galaxy contains 100 billion stars. This number is similar to the number of neurons in human brain (86 billion). The age of universe has been estimated at around 14 billion years as objects as far as 14 billion light years have been observed . The iteration of number at different level is remarkable. If neurons in human brain correspond to a galaxy in the universe and a star in our galaxy, can we postulate the cerebral cortex as the time clock of the universe?

In trees, we do have tree rings by counting which, one can calculate the age of the tree. The subject of Dendrochronology has been developed around this concept. Can we postulate the cerebral cortical neurons as the neuronal marker of the universe's age? Biology is replete with similar concept of recapitulation. Ernst Haeckel gave the hypothesis of “Ontogeny recapitulates phylogeny” after studying embryology. Human embryo grows in stages and appears like a fish, frog, bird and so on. Haeckel claimed that individual passes through evolutionary history in one's own lifetime.

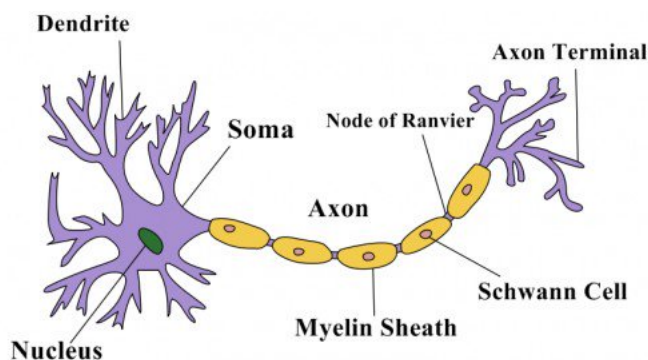
If neuroscience and cosmology converges in future, we may discover similar recapitulation in cortical region. This way the bounded Puruṣa within human body can be seen linked with the cosmic Puruṣa beyond.

Is Sentience an Algorithm?

There has been explosion in research on artificial intelligence. The artificial intelligence is evolving through three stages- Artificial Narrow Intelligence(ANI) to Artificial General Intelligence(AGI) and finally Artificial Super Intelligence(ASI). Mankind is still struggling with building machines with AGI. An AGI will have the general intelligence across the spectrum just like human beings. Scientists are imbuing machines with artificial intelligence by working on the pattern recognition from large number of data and thereafter working out a heuristic tool as an algorithmic process. While intelligence is being considered as the capability of pattern recognition and combination of different algorithms, emotional experiences are being viewed as mere biochemical algorithm. A particular pattern of release and absorption of certain neurotransmitters and chemicals, if can be replicated; scientists are certain that the emotional experience can be manufactured. Even weeping, piloerection in awe

and inspiration has been fully understood as such a biochemical algorithm. AGI will have the ability of recognition of complex patterns like that of human intelligence, but would be faster and greater memory storage. Philosopher Yuval Noah Harari talks about decoupling of intelligence with consciousness. Harari writes in his marvellous book, *Homo Deus*(2017), “ We are now developing new types of non-conscious intelligence that can perform such tasks far better than humans. For all these tasks are based on pattern recognition, and non-conscious algorithms may soon excel human consciousness in recognition patterns.”(*Homo Deus*, pp.361-362)

is the power of self-reflection of idea over itself. Indian philosophers have therefore used the simile of mirror image to describe the mechanism of consciousness. When we do typing, sometimes we keep on typing without being conscious. Similarly, several routine works are being performed by us without being conscious. We breathe, our digestive system, circulatory system, nervous system, muscular system keep on working continuously without being aware. Consciousness is to be aware, the self-awareness. But consciousness is not always pure consciousness. There are levels of consciousness. From coma to superconsciousness, a proper gradient scale of consciousness can be drawn.



If intelligence is algorithmic whereby machines can be intelligent without being conscious, can consciousness be seen as an algorithm? This can be formulated only when one is able to measure sentience.

Marcello Massimini and Giulio Tononi pioneered a technique called “zap and zip” to probe whether someone is conscious or not. This is their consciousness- meter. A coil of wire zaps against the scalp sending magnetic energy into the skull. The electric current induces in the brain excites and inhibits neurons. A network of EEG sensors record the signals. The waves rise and fall and die down. The results are then analyzed by data compression algorithm. More predictable the rhythms are, the more likely the brain is unconscious. The team applied mathematics of probability to measure the Perturbational Complexity Index(PCI). When 102 healthy and 48 responsive but brain-injured patients were zapped and zipped during both state of conscious and unconscious, it was discovered that when the PCI value was above 0.31, the person was conscious and if below this figure, the person was always unconscious. The PCI in normal awake state was always between 0.31 and 0.70 .

Tononi is also the architect of the Integrated information theory(IIT). IIT posits that consciousness is a feature of the universe that emerges with the complexity of the network. Just as Einstein theories that gravitational force is a feature of universe arising from geometry of the space-time; IIT brings in geometry of neuronal arrangements as the cause of consciousness. IIT proposes to measure consciousness by a parameter ϕ , Φ . PCI is an approximation of ϕ , as of now. In general anaesthesia, PCI is down. Many patients with vegetative state has PCI above 0.31 which shows that they are experiencing things around.

IIT finds that higher the information integration a system possess, more ϕ it has, and higher would be the level of consciousness. Cerebellum though has 69 billion neurons is not

integrated internally. The neurons work in grids and in sequence in a linear mode. But, it is cerebral cortex with 16 billion neurons where there are complex integration and feed back loop mechanisms. When a person is in deep sleep, even though he has all the cerebral neurons functioning; he is not conscious since the information integration circuit is broken. Similarly, even a conscious person when is applied anaesthesia, his information integration gets broken and he loses the consciousness. IIT explains how with drugs or alcohol intake, the capacity of integration of information gets affected. IIT predicts that the “spatio-temporal grain of the physical elements specifying consciousness is that yielding the maximum Φ .” Loss and recovery of consciousness is associated merely with this breakdown and recovery of the capacity for information integration. IIT also explains that consciousness cannot be computed, this must be built into the structure of the system. A simulation of nuclear blast does not kill the person. Similarly, even if human brain is simulated and a digital brain created, consciousness cannot emerge. Consciousness is property of a network. IIT postulates that any system-architecture similar to topographically organized areas in human cortex(posterior cortex is where the seat of information integration is guessed), the system would be highly consciousness even if not engaged in intelligent behaviour.

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The Kalacakra Tantra deliberates upon the process by which indestructible quality in different perceptions can be produced. By discussing the aromatics, the text takes the case of the fragrance of musk and how when musk is processed, even a small amount of it can pierce and transfer the fragrance to other concoction. The text describes similar process in astronomy where Rahu pierces the planets through eclipse; alchemy based upon piercing by processed mercury, sexuality where piercing is done by a purified mahamudra; breathing yoga to pierce the nadis with the avadhuti and so on. Finally, the Kalki is such a processed humanized entity which pierces the world afflicted by barbarianism and fill the world with the qualities of dharma that the Kalki possess.

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Will the prophecy in the Kalacakra Tantra about the arising of the Rudra- Kalki in the Age of Algorithm materialize?

Figure 2: List of 36 Masters in Kalacakra Tradition

(Niraj Kumar is a philosopher of Pan-Asianism. He is working on five volume translation and commentary of the Kalacakra Tantra. He can be contacted at kniraj14@yahoo.com)

KAL CHAKRA AND TIME AWARENESS FROM A NEUROLOGICAL PERSPECTIVE .

Neeraj kumar .

Neurons in Human Brain

A neuron has thousands of branches called dendrites. The head and tail both sides have these branching. Then, there are synapses on these dendrites through which the ions are transported and information exchanged. There are 1000-10000 synapses per neuron in human brain. On an average, this number is 7000 synapses per neuron. Synapses are the eyes of neurons through which information is exchanged. The description of Puruṣa in the Veda fits well with structure of neuron. Interestingly, it is these neurons which mediate active consciousness. Can we decipher the intuitive flash of seers through the Puruṣa-Sukta further?

Image 4: Structure of Neuron

Seat of Creative Consciousness: Billions of Neurons

The first verse of the Puruṣa –Sukta gives four numbers- thousand, thousand, thousand and ten. When we multiply all these, we get the number 1010 which is 10 billion. The total number of neurons estimated in human brain is 86 billion. There are 1 billion neurons in spinal cord, 69 billion in cerebellum and 16 billion in cerebral cortex. Recent advances in neuroscience has clearly established that cerebrum is the seat of dynamic consciousness.

Few years back, Nobel Prize winner biologist for discovery of double helix structure of DNA, Francis Crick with Christof Koch(2007) advanced argument for neuronal correlate of consciousness(NCC) defined as minimal neuronal mechanism jointly sufficient for any specific conscious experience. Since then search for “consciousness neurons” has become a challenging exploration.

Even if spinal cord is severed, victims are merely paralyzed in legs, arms and torso and without body sensation. But. They continue to be conscious and experience life in variegated forms. Similarly, even if cerebellum which is involved with motor control, posture, and which possess highest number of neurons is found to be not necessary for conscious experience. Even if the cerebellum gets damaged, a patient does not lose consciousness. Cerebellum is not involved in subjective experience. The conscious experience arises only when neurons in cerebral cortex is excited. Cerebral cortex is the largest part of the brain. There are two hemispheres on right and left joined by a bundle of fibres, corpus callosum, that carries information between the two hemispheres. Each hemisphere has four lobes-frontal, parietal,

temporal(side) and occipital(back side). Thus, the cerebrum comprises of 8 lobes containing a total of 16 billion neurons.

The number interpreted from the first verse of the Puruṣa –Sukta, i.e. 10 billion is very close to 16 billion.

Since the set of 16 billion neurons is the seat of active consciousness, does the cortical neurons relate to the outer world in anyway?

Cortical Neurons and Cosmic History

There are 100 billion galaxies estimated to be scattered in our universe. We again find that our galaxy contains 100 billion stars. This number is similar to the number of neurons in human brain (86 billion). The age of universe has been estimated at around 14 billion years as objects as far as 14 billion light years have been observed . The iteration of number at different level is remarkable. If neurons in human brain correspond to a galaxy in the universe and a star in our galaxy, can we postulate the cerebral cortex as the time clock of the universe?

In trees, we do have tree rings by counting which, one can calculate the age of the tree. The subject of Dendrochronology has been developed around this concept. Can we postulate the cerebral cortical neurons as the neuronal marker of the universe's age? Biology is replete with similar concept of recapitulation. Ernst Haeckel gave the hypothesis of “Ontogeny recapitulates phylogeny” after studying embryology. Human embryo grows in stages and appears like a fish, frog, bird and so on. Haeckel claimed that individual passes through evolutionary history in one's own lifetime.

If neuroscience and cosmology converges in future, we may discover similar recapitulation in cortical region. This way the bounded Puruṣa within human body can be seen linked with the cosmic Puruṣa beyond.

Is Sentience an Algorithm?

There has been explosion in research on artificial intelligence. The artificial intelligence is evolving through three stages- Artificial Narrow Intelligence(ANI) to Artificial General Intelligence(AGI) and finally Artificial Super Intelligence(ASI). Mankind is still struggling with building machines with AGI. An AGI will have the general intelligence across the spectrum just like human beings. Scientists are imbuing machines with artificial intelligence by working on the pattern recognition from large number of data and thereafter working out a heuristic tool as an algorithmic process. While intelligence is being considered as the capability of pattern recognition and combination of different algorithms, emotional experiences are being viewed as mere biochemical algorithm. A particular pattern of release and absorption of certain neurotransmitters and chemicals, if can be replicated; scientists are certain that the emotional experience can be manufactured. Even weeping, piloerection in awe and inspiration has been fully understood as such a biochemical algorithm. AGI will have the ability of recognition of complex patterns like that of human intelligence, but would be faster and greater memory storage. Philosopher Yuval Noah Harari talks about decoupling of intelligence with consciousness. Harari writes in his marvellous book, *Homo Deus*(2017), “ We are now developing new types of non-conscious intelligence that can perform such tasks far better than humans. For all these tasks are based on pattern recognition, and non-

conscious algorithms may soon excel human consciousness in recognition patterns.”(Homo Deus, pp.361-362)

is the power of self-reflection of idea over itself. Indian philosophers have therefore used the simile of mirror image to describe the mechanism of consciousness. When we do typing, sometimes we keep on typing without being conscious. Similarly, several routine works are being performed by us without being conscious. We breathe, our digestive system, circulatory system, nervous system, muscular system keep on working continuously without being aware. Consciousness is to be aware, the self-awareness. But consciousness is not always pure consciousness. There are levels of consciousness. From coma to superconsciousness, a proper gradient scale of consciousness can be drawn.

Table 1: Content of Consciousness

Content of Consciousness

Samadhi

REM Awareness of Dhyana

Conscious Wakefulness

Day Dreaming

Drowsiness

Light Sleep

REM Sleep(Rapid Eye Movement)

Deep Sleep

Sleep walking, seizure, vegetative State

General Anaesthesia

Coma

Death

If intelligence is algorithmic whereby machines can be intelligent without being conscious, can consciousness be seen as an algorithm? This can be formulated only when one is able to measure sentience.

Marcello Massimini and Giulio Tononi pioneered a technique called “zap and zip” to probe whether someone is conscious or not. This is their consciousness-meter. A coil of wire zaps against the scalp sending magnetic energy into the skull. The electric current induces in the brain excites and inhibits neurons. A network of EEG sensors record the signals. The waves rise and fall and die down. The results are then analyzed by data compression algorithm. More predictable the rhythms are, the more likely the brain is unconscious. The team applied mathematics of probability to measure the Perturbational Complexity Index (PCI). When 102 healthy and 48 responsive but brain-injured patients were zapped and zipped during both state of conscious and unconscious, it was discovered that when the PCI value was above 0.31, the person was conscious and if below this figure, the person was always unconscious. The PCI in normal awake state was always between 0.31 and 0.70.

Tononi is also the architect of the Integrated information theory (IIT). IIT posits that consciousness is a feature of the universe that emerges with the complexity of the network. Just as Einstein theories that gravitational force is a feature of universe arising from geometry of the space-time; IIT brings in geometry of neuronal arrangements as the cause of consciousness. IIT proposes to measure consciousness by a parameter ϕ , Φ . PCI is an approximation of ϕ , as of now. In general anaesthesia, PCI is down. Many patients with vegetative state has PCI above 0.31 which shows that they are experiencing things around.

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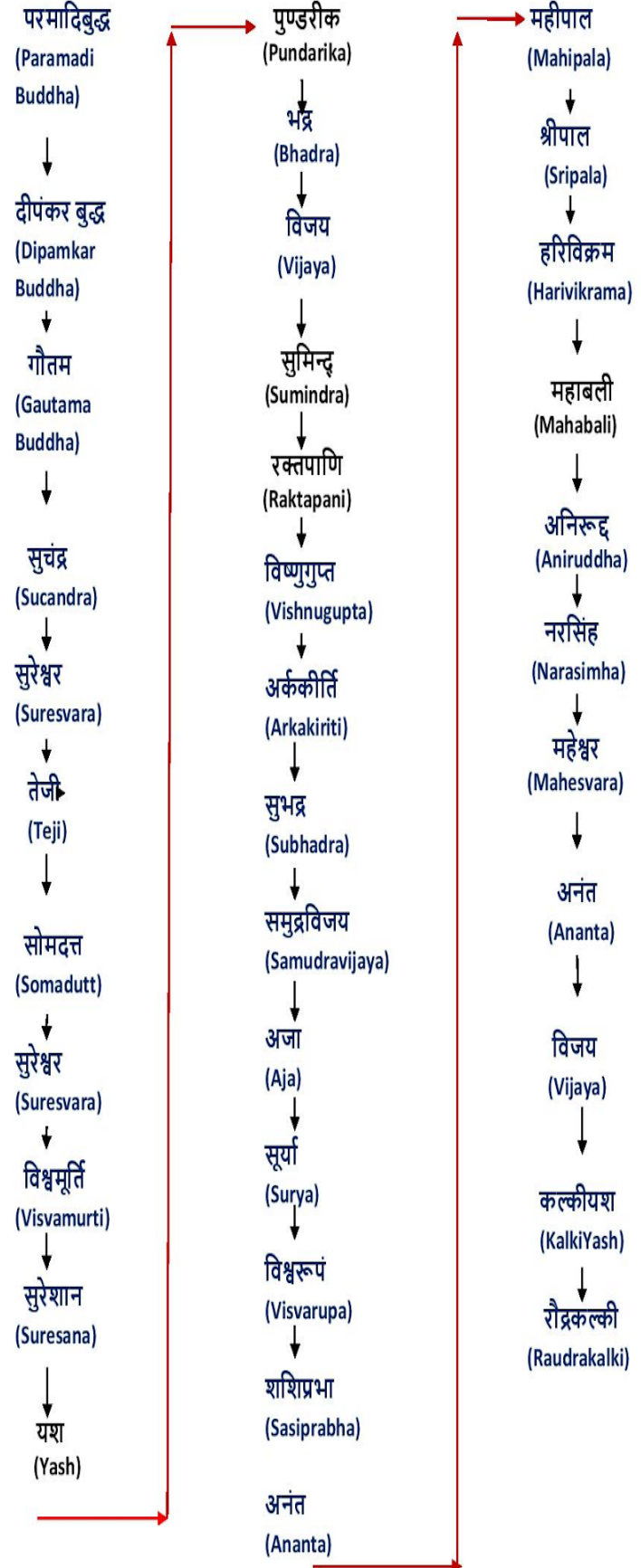
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Poetry.

Everywhere Rainbows.

~ Dr.Ambika Talwar

everywhere rainbows

arc in purple skies

cosmos glows

beauty herein lies

dew drops along spinal cord

whole beings wander

in the dark of time

where light swishes her tail

winding spreading

weaving my skin

blanket of dreams such scapes

everywhere is home..

woven cosmic sounds

contain our seeing each other

dressed in rainbow...

clear as glass
we glide on midori lawn
soaked with sunlight...

Dr Ambika Talwar teaches creative arts in the US.

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